Nonequilibrium odd-even relaxation effect in tight binding models where Fermi surface is near the van Hove singularity

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Contents

Contents 1

1 Calculation of the collision kernel 2

1.1 Linearization of the collision integral 2

1.2 Preliminary simplification of the Fermi factor 3

1.3 Kinematic constraints 4

CHAPTER 1

Calculation of the collision kernel

1.1 Linearization of the collision integral

The Boltzmann equation is

$$\frac{\mathrm{d}f}{\mathrm{d}t} + \mathbf{v} \cdot \frac{\mathrm{d}f}{\mathrm{d}\mathbf{r}} + \mathbf{F} \cdot \frac{\mathrm{d}f}{\mathrm{d}\hbar\mathbf{p}} = \mathcal{J}[f] \tag{1.1}$$

where $f(t, \mathbf{x}, \mathbf{p})$ is the one particle density function, and $\mathcal{J}$, the collision integral given by

$$\mathcal{J}[f] = -\frac{1}{(2\pi)^6} \int\!\!\ldots\!\!\int \mathrm{d}\mathbf{p}_2 \, \mathrm{d}\mathbf{p}_1' \, \mathrm{d}\mathbf{p}_2' W(\mathbf{p}_1', \mathbf{p}_2' | \mathbf{p}_1, \mathbf{p}_2) \mathcal{B} \tag{1.2}$$

where

$$\begin{aligned} \mathcal{B} = \big[& f(\mathbf{p}_1) f(\mathbf{p}_2) (1 - f(\mathbf{p}_1')) (1 - f(\mathbf{p}_2')) \\ & - f(\mathbf{p}_1') f(\mathbf{p}_2') (1 - f(\mathbf{p}_1)) (1 - f(\mathbf{p}_2)) \big] \end{aligned} \tag{1.3}$$

This integral is non-linear and is hard to solve. Hence in this section, we shall simplify the integral by linearizing it w.r.t. f.

From here, we shall write $f_i = f(\mathbf{p}_i)$ and $f_{0,i} = f(\mathbf{p}_i)$. Primed f_i' are understood as $f(\mathbf{p}_i')$.

Consider the taylor expansion of $\mathcal{B}$ at f_0:

$$\mathcal{B} = \mathcal{B}|_{f_0} + \mathcal{B}_{\text{linear}} + \mathcal{B}_{\text{non-linear}}. \tag{1.4}$$

The linear part can be expanded w.r.t. each of the f_i's to get

$$\mathcal{B}_{\text{linear}} = \delta f_1 \left(\frac{\partial \mathcal{B}}{\partial f_1}\right)_{f_0} + \delta f_2 \left(\frac{\partial \mathcal{B}}{\partial f_2}\right)_{f_0} + \delta f_1' \left(\frac{\partial \mathcal{B}}{\partial f_1'}\right)_{f_0} + \delta f_2' \left(\frac{\partial \mathcal{B}}{\partial f_2'}\right)_{f_0} \tag{1.5}$$

1.2 Preliminary simplification of the Fermi factor

Let $a := \beta(\varepsilon(\mathbf{p}_1) - \mu)$ and $b := \beta(\varepsilon(\mathbf{p}_2) - \mu)$, then

$$f_0(\mathbf{p}_1) f_0(\mathbf{p}_2) = \frac{1}{(1+\mathrm{e}^a)(1+\mathrm{e}^b)} \tag{1.6}$$

$$= \frac{1}{1+\mathrm{e}^a+\mathrm{e}^b+\mathrm{e}^{a+b}} \tag{1.7}$$

$$= \frac{1}{\mathrm{e}^{\frac{a+b}{2}}\left(\mathrm{e}^{-\frac{a+b}{2}}+\mathrm{e}^{\frac{a+b}{2}}+\mathrm{e}^{\frac{a-b}{2}}+\mathrm{e}^{-\frac{a-b}{2}}\right)} \tag{1.8}$$

$$= \frac{1}{2\mathrm{e}^{\frac{a+b}{2}}\left[\frac{1}{2}(\mathrm{e}^{-\frac{a+b}{2}}+\mathrm{e}^{\frac{a+b}{2}})+\frac{1}{2}\left(\mathrm{e}^{\frac{a-b}{2}}+\mathrm{e}^{-\frac{a-b}{2}}\right)\right]} \tag{1.9}$$

$$= \frac{1}{2\mathrm{e}^{\frac{a+b}{2}}\left[\cosh(\frac{a+b}{2})+\cosh(\frac{a-b}{2})\right]} \tag{1.10}$$

Similarly, let $a' := \beta(\varepsilon(\mathbf{p}_1') - \mu)$ and $b' := \beta(\varepsilon(\mathbf{p}_2') - \mu)$, then

$$(1-f_0(\mathbf{p}_1'))(1-f_0(\mathbf{p}_2')) = \left(\frac{\mathrm{e}^{a'}}{1+\mathrm{e}^{a'}}\right)\left(\frac{\mathrm{e}^{b'}}{1+\mathrm{e}^{b'}}\right) \tag{1.11}$$

$$= \frac{\mathrm{e}^{a'+b'}}{(1+\mathrm{e}^{a'})(1+\mathrm{e}^{b'})} \tag{1.12}$$

$$= \frac{\mathrm{e}^{a'+b'}}{2\mathrm{e}^{\frac{a'+b'}{2}}\left[\cosh(\frac{a'+b'}{2})+\cosh(\frac{a'-b'}{2})\right]} \tag{1.13}$$

Together, we get that

$$F_{121'2'} = \frac{1}{2\mathrm{e}^{\frac{a+b}{2}}\left[\cosh(\frac{a+b}{2})+\cosh(\frac{a-b}{2})\right]} \cdot \frac{\mathrm{e}^{a'+b'}}{2\mathrm{e}^{\frac{a'+b'}{2}}\left[\cosh(\frac{a'+b'}{2})+\cosh(\frac{a'-b'}{2})\right]}$$

By energy conservation, $a' + b' = a + b$. Hence, $F_{121'2'}$ evaluates to

$$\frac{1}{4\Big[\cosh\xi + \cosh\Big(\frac{\beta}{2}(\varepsilon(\mathbf{p}_1) - \varepsilon(\mathbf{p}_2))\Big)\Big]\Big[\cosh\xi + \cosh\Big(\frac{\beta}{2}(\varepsilon(\mathbf{p}_1') - \varepsilon(\mathbf{p}_2'))\Big)\Big]} \tag{1.14}$$

where $\xi := \dfrac{\beta}{2}(\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - 2\mu)$.

1.3 Kinematic constraints

The kinematic constraints includes the momentum and energy conservation which will be imposed into the integral via dirac deltas:

$$\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_1' - \mathbf{p}_2' = 0, \tag{1.15}$$

$$\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - \varepsilon(\mathbf{p}_1') - \varepsilon(\mathbf{p}_2') = 0. \tag{1.16}$$

To evaluate these dirac deltas, we turn into the center of momentum and relative momentum coordinates defined as:

$$\mathbf{S} = \mathbf{p}_1 + \mathbf{p}_2 \qquad \mathbf{p}_1 = \frac{1}{2}(\mathbf{S} + \mathbf{D}) \tag{1.17}$$

$$\mathbf{D} = \mathbf{p}_1 - \mathbf{p}_2 \qquad \mathbf{p}_2 = \frac{1}{2}(\mathbf{S} - \mathbf{D}). \tag{1.18}$$

The same scheme is used for the primed variables.

Here are then some useful identities that will be used later on. Firstly, the sum of the square of momentas in each axes:

$$p_{1x}^2 + p_{2x}^2 = \left(\frac{1}{2}(S_x + D_x)\right)^2 + \left(\frac{1}{2}(S_x - D_x)\right)^2 \tag{1.19}$$

$$= \frac{1}{4}\left(S_x^2 + 2S_xD_x + D_x^2 + S_x^2 - 2S_xD_x + D_x^2\right) \tag{1.20}$$

$$= \frac{1}{4}(2S_x^2 + 2D_x^2) = \frac{1}{2}(S_x^2 + D_x^2) \tag{1.21}$$

In a similar manner,

$$p_{1y}^2 + p_{2y}^2 = \frac{1}{2}(S_y^2 + D_y^2). \tag{1.22}$$

Secondly, the difference of momentas squared:

$$p_{1x}^2 - p_{2x}^2 = \left(\frac{1}{2}(S_x + D_x)\right)^2 - \left(\frac{1}{2}(S_x - D_x)\right)^2 \tag{1.23}$$

$$= \frac{1}{4}(S_x^2 + 2S_xD_x + D_x^2 - S_x^2 + 2S_xD_x - D_x^2) \tag{1.24}$$

$$= \frac{1}{4}(4S_xD_x) = S_xD_x. \tag{1.25}$$

Similarly,

$$p_{2x}^2 - p_{2y}^2 = S_yD_y. \tag{1.26}$$

Hence,

$$\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) = \frac{\hbar^2}{2m}(p_{1x}^2 - p_{1y}^2 + p_{2x}^2 - p_{2y}^2) \tag{1.27}$$

$$= \frac{\hbar^2}{2m}\Big((p_{1x}^2 + p_{2x}^2) - (p_{1y}^2 + p_{2y}^2)\Big) \tag{1.28}$$

$$= \frac{\hbar^2}{4m}(S_x^2 + D_x^2 - S_y^2 - D_y^2). \tag{1.29}$$

By detailed balance, $\varepsilon(\mathbf{p}_1') + \varepsilon(\mathbf{p}_2')$ is also equal to this. For the difference in energy,

$$\varepsilon(\mathbf{p}_1) - \varepsilon(\mathbf{p}_2) = \frac{\hbar^2}{2m}(p_{1x}^2 - p_{1y}^2 - (p_{2x}^2 - p_{2y}^2)) \tag{1.30}$$

$$= \frac{\hbar^2}{2m}(p_{1x}^2 - p_{2x}^2 - (p_{1y}^2 - p_{2y}^2)) \tag{1.31}$$

$$= \frac{\hbar^2}{2m}(S_xD_x - S_yD_y). \tag{1.32}$$

Similarly,

$$\varepsilon(\mathbf{p}_1') - \varepsilon(\mathbf{p}_2') = \frac{h^2}{2m}(S_x'D_x' - S_y'D_y'). \tag{1.33}$$

For ease of reading, we shall restate the integral:

$$\mathcal{J}[f] = -\frac{1}{(2\pi)^6}\int\!\!\ldots\!\!\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1'\,\mathrm{d}\mathbf{p}_2' \times \frac{(2\pi)^3}{\hbar}|V|^2 \tag{1.34}$$

$$\times\,\delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_1' - \mathbf{p}_2')\,\delta(\varepsilon(\mathbf{p}_1) + \varepsilon(\mathbf{p}_2) - \varepsilon(\mathbf{p}_1') - \varepsilon(\mathbf{p}_2'))$$

$$\times\, F_{121'2'}\big[\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')\big].$$

$$= -\frac{|V|^2}{8\pi^3\hbar}\int\!\!\ldots\!\!\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1'\,\mathrm{d}\mathbf{p}_2' \tag{1.35}$$
$$\times\,\delta(\mathbf{p}_1+\mathbf{p}_2-\mathbf{p}_1'-\mathbf{p}_2')\,\delta(\varepsilon(\mathbf{p}_1)+\varepsilon(\mathbf{p}_2)-\varepsilon(\mathbf{p}_1')-\varepsilon(\mathbf{p}_2'))$$
$$\times\,F_{121'2'}\big[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\big].$$

Performing the change of coordinates, we get

$$\mathcal{J}[f] = -\frac{|V|^2}{8\pi^3\hbar}\int\!\!\ldots\!\!\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1'\,\mathrm{d}\mathbf{p}_2'\times\delta(\mathbf{S}-\mathbf{S}') \tag{1.36}$$
$$\times\,\frac{4m}{\hbar^2}\delta(S_x^2+D_x^2-S_y^2-D_y^2-S_x'^2-D_x'^2+S_y'^2+D_y'^2)$$
$$\times\,F_{121'2'}\big[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\big].$$
$$= -\frac{m|V|^2}{2\pi^3\hbar^3}\int\!\!\ldots\!\!\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{p}_1'\,\mathrm{d}\mathbf{p}_2'\times\delta(\mathbf{S}-\mathbf{S}') \tag{1.37}$$
$$\times\,\delta(S_x^2+D_x^2-S_y^2-D_y^2-S_x'^2-D_x'^2+S_y'^2+D_y'^2)$$
$$\times\,F_{121'2'}\big[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\big].$$

We then change the measure from $(\mathbf{p}_1', \mathbf{p}_2')$ to $(\mathbf{S}', \mathbf{D}')$. The jacobian can be calculated as follows:

$$\begin{vmatrix} \frac{\partial S_x'}{\partial p_{1x}'} & \frac{\partial S_x'}{\partial p_{1y}'} & \frac{\partial S_x'}{\partial p_{2x}'} & \frac{\partial S_x'}{\partial p_{2y}'} \\ \frac{\partial S_y'}{\partial p_{1x}'} & \frac{\partial S_y'}{\partial p_{1y}'} & \frac{\partial S_y'}{\partial p_{2x}'} & \frac{\partial S_y'}{\partial p_{2y}'} \\ \frac{\partial D_x'}{\partial p_{1x}'} & \frac{\partial D_x'}{\partial p_{1y}'} & \frac{\partial D_x'}{\partial p_{2x}'} & \frac{\partial D_x'}{\partial p_{2y}'} \\ \frac{\partial D_y'}{\partial p_{1x}'} & \frac{\partial D_y'}{\partial p_{1y}'} & \frac{\partial D_y'}{\partial p_{2x}'} & \frac{\partial D_y'}{\partial p_{2y}'} \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{vmatrix} = 4. \tag{1.38}$$

Hence,

$$\mathrm{d}\mathbf{p}_1'\,\mathrm{d}\mathbf{p}_2' = \left|\frac{\partial\left(p_{1x}', p_{1y}', p_{2x}', p_{2y}'\right)}{\partial\left(S_x', S_y', D_x', D_y'\right)}\right|\mathrm{d}\mathbf{S}\,\mathrm{d}\mathbf{D} = \frac{1}{4}\,\mathrm{d}\mathbf{S}\,\mathrm{d}\mathbf{D}, \tag{1.39}$$

$$\mathcal{J}[f] = \frac{m|V|^2}{2\pi^3\hbar^3}\int\!\!\ldots\!\!\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{S}'\,\mathrm{d}\mathbf{D}'\times \tag{1.40}$$
$$\frac{1}{4}\delta(\mathbf{S}-\mathbf{S}')\delta(S_x^2+D_x^2-S_y^2-D_y^2-S_x'^2-D_x'^2+S_y'^2+D_y'^2)$$
$$\times\,F_{121'2}\big[\psi(\mathbf{p}_1)+\psi(\mathbf{p}_2)-\psi(\mathbf{p}_1')-\psi(\mathbf{p}_2')\big]$$

$$= \frac{m|V|^2}{8\pi^3\hbar^3}\int\!\!\ldots\!\int \mathrm{d}\mathbf{p}_2\,\mathrm{d}\mathbf{D}' \times \delta\Big((D_x^2 - D_x'^2) - (D_y^2 - D_y'^2)\Big) \quad . \tag{1.41}$$
$$\times\, F_{121'2}\big[\psi(\mathbf{p}_1) + \psi(\mathbf{p}_2) - \psi(\mathbf{p}_1') - \psi(\mathbf{p}_2')\big].$$

Under this coordinate, the Fermi factor becomes

$$\frac{1}{4\Big[\cosh\xi + \cosh\Big(\frac{\beta\hbar^2}{4m}(S_xD_x - S_yD_y)\Big)\Big]\Big[\cosh\xi + \cosh\Big(\frac{\beta\hbar^2}{4m}(S_x'D_x' - S_y'D_y')\Big)\Big]} \tag{1.42}$$

where ξ now becomes $\dfrac{\beta\hbar^2}{8m}(S_x^2 + D_x^2 - S_y^2 - D_y^2) - \beta\mu$.